

Table 3

Overall error in wall superheat of apparent boiling curves (heat loss simulation vs experimental values).

Device no.	$R_{a,low}$ (μm)	RMS error (K)
1	0.263	0.843
2	0.622	0.568
3	1.396	0.746
4	4.25	0.375
5	7.51	0.332
6	4.40	0.531
7	3.73	0.470

determine the contact angle. In the differencing operation, intensity values between 0 and 1 of each pixel in the reference image were subtracted from the corresponding values in the droplet image and the absolute value of the result was increased in contrast by 200%. Forty or more samples were analyzed for each surface (except for surface 5, for which 22 samples were analyzed), resulting in uncertainties between 0.4° and 1.0° . The contact angles decreased almost monotonically with measured R_a values.

5. Data reduction

5.1. Heat loss calculation

In constructing boiling curves, the temperature data (abscissa) could be easily calculated from the calibration curves described in Section 4.1. The boiling heat flux data (ordinate), however, could not be obtained in a similar direct manner. The power dissipation,